1 Definitions

For a function $f : [n] \to \mathbb{R}$ we say a pair $(i, i+1)$ for i in $[n-1]$ is

1. a *good pair* if there exists an optimal LIS that uses both elements in the pair
2. a *bad pair* if no optimal LIS includes both elements in the pair
3. a *perfect pair* is a pair that must be used in all optimal LIS

Note that all $(j, j+1)$ such that $f(j) > f(j+1)$ are "bad pairs". Also note that all perfect pairs are also good pairs

2 $1 + \delta$ assuming $\varepsilon_f < 0.5$

Theorem 2.1. *For a fixed delta, if the true ε_f is less than 0.5, Using distance-separation(f, ε, δ), the decision problem of if f is within $\varepsilon \pm \delta$ is solvable in polylog time.*

proof of theorem 2.1. First, run DistApprox □

3 good pair analysis

3.1 fraction of good pairs

Theorem 3.1. *At least half of all increasing pairs are good pairs*

3.2 padding good pairs

Consider the construction of a new function $f_p : [n+1] \to \mathbb{R}$ defined as

$$f_p(x) = \begin{cases} f(x), & x \leq i \\ \frac{f(i)+f(i+1)}{2}, & x = i+1 \\ f(x+1), & x \geq i+2 \end{cases} \tag{1}$$

Theorem 3.2. *given a function f and a good pair of f $(i, i+1)$, f_p forces $(i, i+1), (i+1, i+2)$ to both be perfect pairs in the new function f_p. Additionally, f_p has a distance to monotonicity of $\frac{\varepsilon_f n+1}{n+1}$*